

GeoChemFoam short course 2026

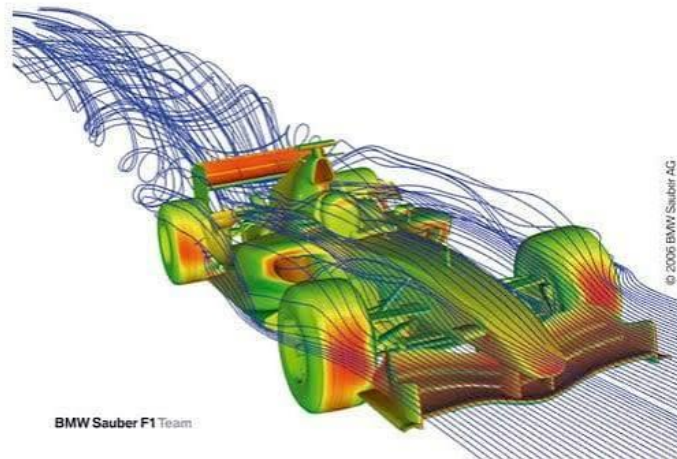
Dr Julien Maes, Dr Jacqueline Mifsud

Agenda

- 9:30-10:00: Introduction to GeoChemFoam
- 10:00-11:00: From a micro-CT image to a GeoChemFoam case
- 11:00-11:15: Coffee break
- 11:15-12:15: Flow and permeability calculation
- 12:15-13:30: Lunch break
- 13:30-14:00: GeoChemFoam/Phreeqc coupling: how does it work
- 14:00-15:00: Reactive transport: Nickel (de)sorption on Na-Illite surfaces
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- 16:15-16:45: Conclusion and discussion

What is OpenFOAM

An open-source C++ toolbox for building and running computational fluid dynamics (CFD) solvers.



PHYSICS

Navier-Stokes equations

conservation of mass + momentum

NUMERICS

Finite-volume discretisation

mesh $\rightarrow$ algebraic equations $\rightarrow$ iterative solution

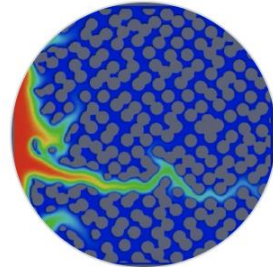
OUTPUT

Fields in space and time

velocity U • pressure p • species C_i

Not typically for reservoir scale (Darcy) but for pore-scale (NS)

What is GeoChemFoam?



GeoChemFoam

Follow

147 followers · 2 following

Edinburgh, UK

Achievements



Block or report user

GeoChemFoam / README.md

Hi there 🙌, welcome to The GeoChemFoam Project github! Here you will find links to the [source code](#), [precompiled dockers](#), [a users wiki with tutorials, publications](#), [a discussion forum](#), a [YouTube channel](#) and more!

The GeoChemFoam Project source code is hosted here on [github](#). A precompiled docker of the latest version is also available on the [docker hub](#).

The GeochemFoam Project is the world's most advanced open source pore-scale numerical simulator based on OpenFOAM founded at the Institute of GeoEnergy Engineering at Heriot-Watt University in 2019 by [Dr Julien Maes](#) and [Dr Hannah Menke](#). GeoChemFoam is made specifically for researching pore-scale processes related to the energy transition and our NetZero carbon future. Below is a list of current capabilities, but **we are always adding new stuff and we welcome contributions from the community**. Drop us a line at GeoChemFoam@hw.ac.uk.

Project Partners:

Dr Gavin Pringle ([EPCC, University of Edinburgh](#)) Contribution to GeoChemFoam: developed the parallel meshing tool introduced in GeoChemFoam 5.2 and maintains the GeoChemFoam module on [ARCHER2](#).

Contributors:

Dr Catherine Spurin ([Stanford University](#)) Contribution to GeoChemFoam: developed the [jupyter notebook workflow for image processing](#) and helped us adapt it for use inside the GeoChemFoam 5.2 docker.

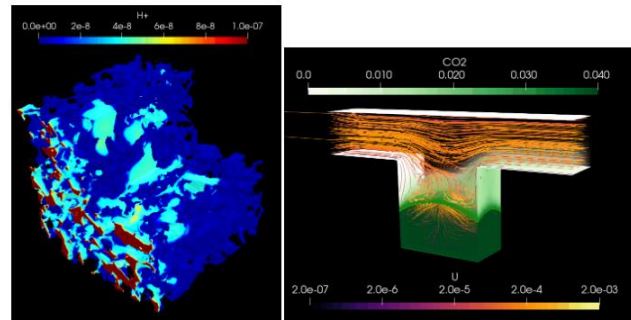
Dr Shaina Kelly ([Kelly Lab, Columbia University](#)) Contribution to GeoChemFoam: built the Singularity container used for running GeoChemFoam 5.1 on HPC systems.

Dr Alexandros Patsoukis Dimou ([University of Edinburgh](#)) Contribution to GeoChemFoam: developed *scalarTransportDBSFoam* as part of his PhD project.

Dr Saideep Pavuluri ([Heriot-Watt University](#)) Contribution to GeoChemFoam: developed *InterGCFoam* as part of his PhD project.

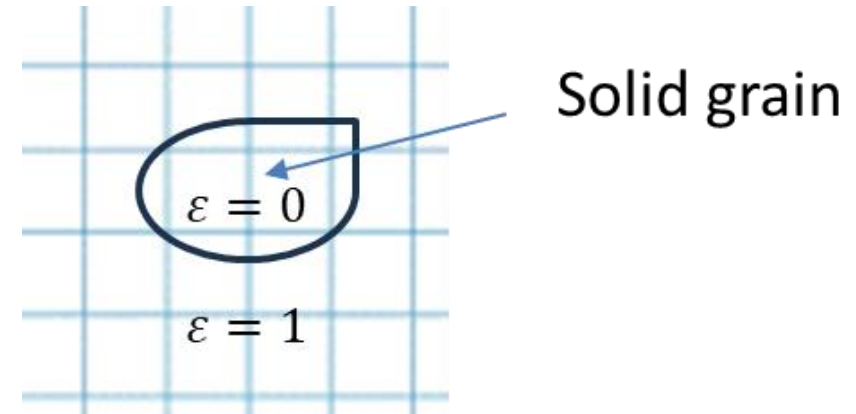
Current capabilities:

- Multiphase Reactive Transport (e.g. [scCO2 dissolving in brine](#))

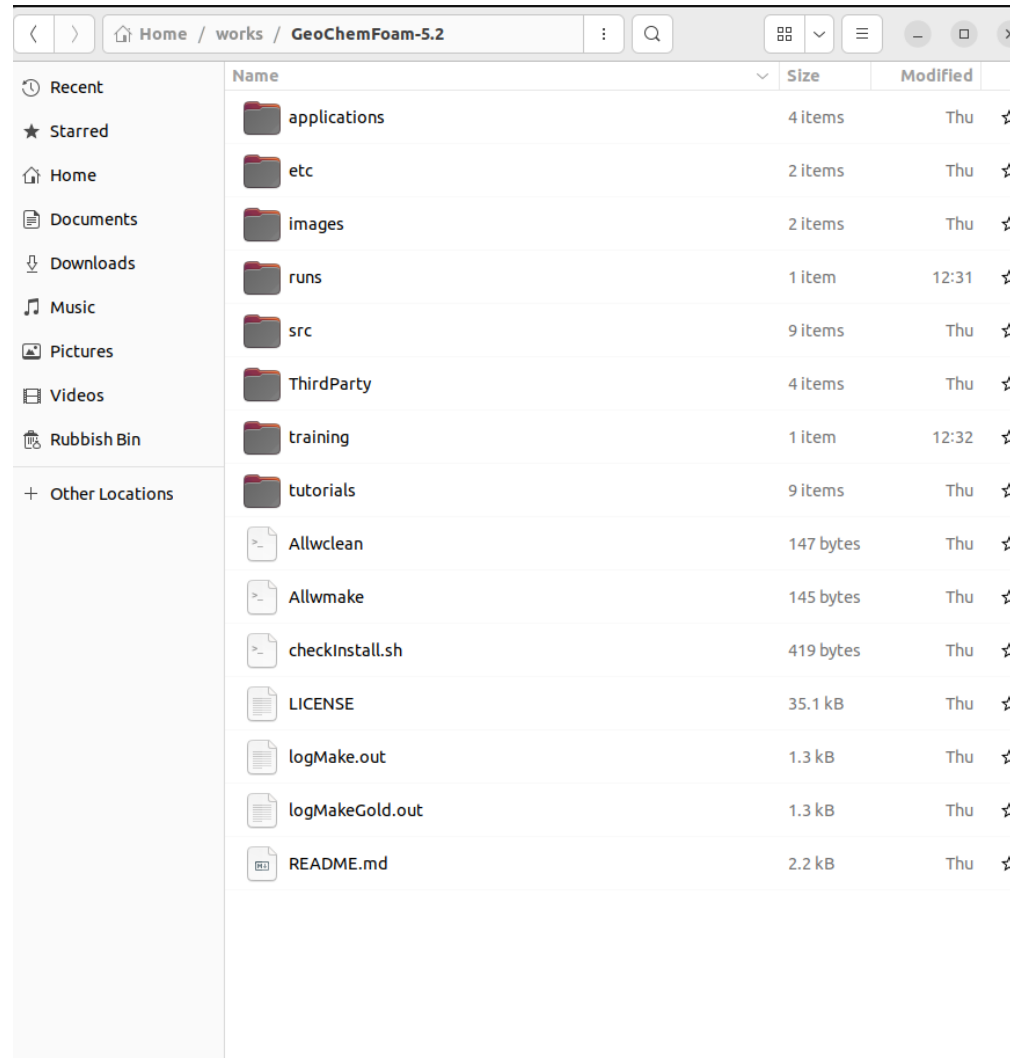


GeoChemFoam core method

- Flow resolved using the Darcy-Brinkman-Stokes equations
- Darcy in porous part, Stokes in resolved pores
- Solid penalised by setting low porosity low permeability
- Advantage: microporosity, pore/solid interaction, simple mesh



Tour of the GeoChemFoam source code

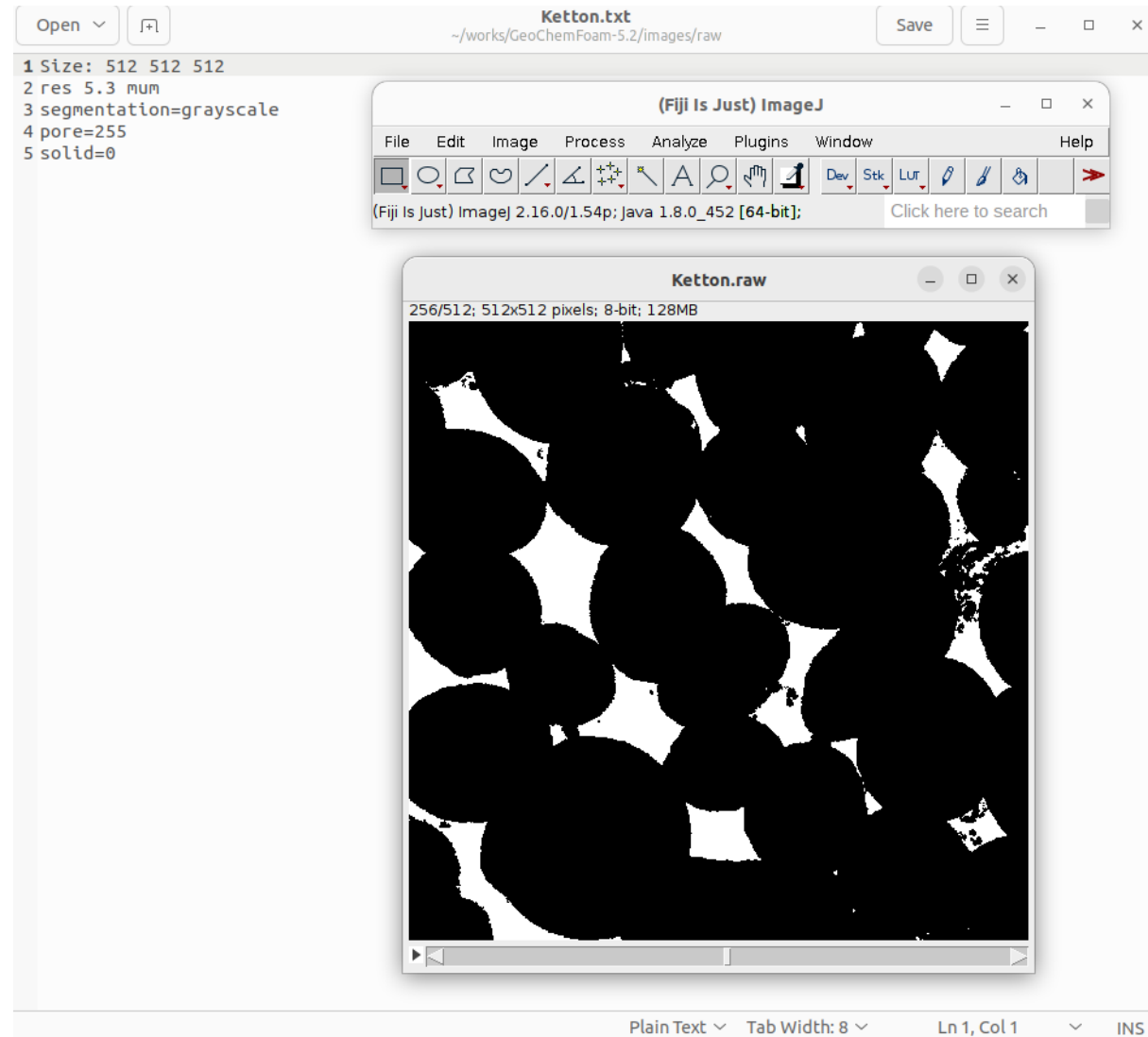


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Micro-CT image

- ImageJ



Create the test case and open the createMesh script

```

julien@ju
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs$ cp -r ../tutorials/incompressible/simpleDBSFoam/est_subvol/ Ketton
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs$ cd Ketton/
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ ls
constant  createMesh.sh  deleteAll.sh  deleteFlow.sh  initCaseFlow.sh  Paraview_Viewer.foam  processFlow.sh  runAll.sh  runCaseFlow.sh  runSmoothSolidSurface.sh  system
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ vi createMesh.sh

```

```

#!/bin/bash
##### USERS INPUT #####
#Define Image name
#image_name="Est_3phase500x500x500micron"
#image_name="401x401x401micron"
#image_name="Bentheimer400-5mm"
#image_name="W112_2_4"
#Image directory location ($PWD if current directory)
dir="$PWD/001/raw"
#Choose Image format
format="raw"
#Choose if the image is compressed or not
compressed="yes"
# Define Image dimensions
x_dim=500
y_dim=500
z_dim=500
#segmentation="grayscale"
#Values of solid, pore, and minimum porosity value for the solid phase (note: if the image contains solid voxels, this CANNOT be 0)
#pores_value=0
#solid_value=255
#pore_min=0.0001
segmentation="phases"
#Values of solid and pore
pores_value=1
solid_value=3
#Define the labels of the phases
phases=(1,2,3)
#Define the porosity of each phase, note that the porosity of the solid phase CANNOT be 0, default to 0.0001
micro_por=(1,0.35,0.0001)
# Define cropping parameters
x_min=100
x_max=300
y_min=100
y_max=300
z_min=100
z_max=300
#padding for inlet/outlet
#This adds voxels in the image directly
padding=4
# define resolution (n)
res=0.000004
# number of cells of initial mesh
# They need to be a factor of the number of voxels in the image
n_x=50
n_y=50
n_z=50
#Mesh refinement level
localRefinement="true"
#Number of layers around interface
layer=2
nlevels=2
refinedStokes=0
refinedSolid=0
dimension="3D"
#dimension="2D"
# Flow direction 0, 1 or 2
direction=0
# Number of processors in each direction
nProc=2
nProc=2
nProc=2
##### END OF USER INPUT #####
if [ $format != "raw" ]
then
    echo "ERROR: only raw format is implemented for this solver"
    exit
fi
if [ "${dimension}" == "2D" ]
then
    if [ $direction == 2 ]
    then
        echo "ERROR: direction cannot be 2 in 2D simulations"
    fi
fi

```

Open the createMesh.sh with your favorite text editor (vi, nano, gedit, notepad...)

We need to change this file according to the simulation we will run, using the Ketton.txt file and the mesh we want to create – this is presented in the next slides

Modify the createMesh.sh script

```
##### USERS INPUT #####
#Define image name
Image_name="Ketton"
#Image_name="4StraightFractures"
#Image_name="Bentheimer400-5mum"
#Image_name="HM12_2_4"

#Image directory location ($PWD if current directory)
dir="$GCFOAM_IMG/raw"

#Choose image format
format='raw'

#Choose if the image is compressed or not
compressed='yes'

# Define image dimensions
x_dim=512
y_dim=512
z_dim=512

segmentation='grayscale'
#Values of solid, pore, and minimum porosity value for the solid phase (note: if the image contains solid voxels, this CANNOT be 0)
pores_value=255
solid_value=0
eps_min=0.0001

#segmentation='phases'
#Values of solid and pore
#pores_value=1
#solid_value=3

#define the labels of the phases
#phases=(1,2,3)

#define the porosity of each phase, note that the porosity of the solid phase CANNOT be 0, default to 0.0001
#inlet_por=(1,0.35,0.0001)

# Define cropping parameters
x_min=0
x_max=256
y_min=0
y_max=256
z_min=0
z_max=256

#padding for inlet/outlet
#This adds voxels in the image directly
padWidth=4
```

Name, location. Note that our image is compressed

Dimension and segmentation properties
For a grayscale image the porosity follows: $\text{eps} = (\text{value-solid})/(\text{pores-solid})$
Rescale so eps not 0. Comment the "phases" properties

Size of the image and pad. Pad creates an injection zone and a production zone on each side

Modify the createMesh.sh script

```
# define resolution (m)  
res=0.0000053
```

Image resolution

```
# number of cells of initial mesh  
## they need to be a factor of the number of voxels in the image  
n_x=64  
n_y=64  
n_z=64  
  
#Mesh refinement level  
localRefinement='true'  
#number of layers around interface  
layer=2  
nlevel=2  
refineStokes=0  
refineSolid=0
```

Mesh: This starts with a coarse grid of 64x64x64 (resolution 4x5.3 microns). Then we create a 2-level mesh by refining cells at the interface (resolution 2x5.3 microns)

```
dimension="3D"  
#dimension="2D"  
# flow direction 0, 1 or 2  
direction=0
```

Dimension and flow direction (correspond to x, y and z)

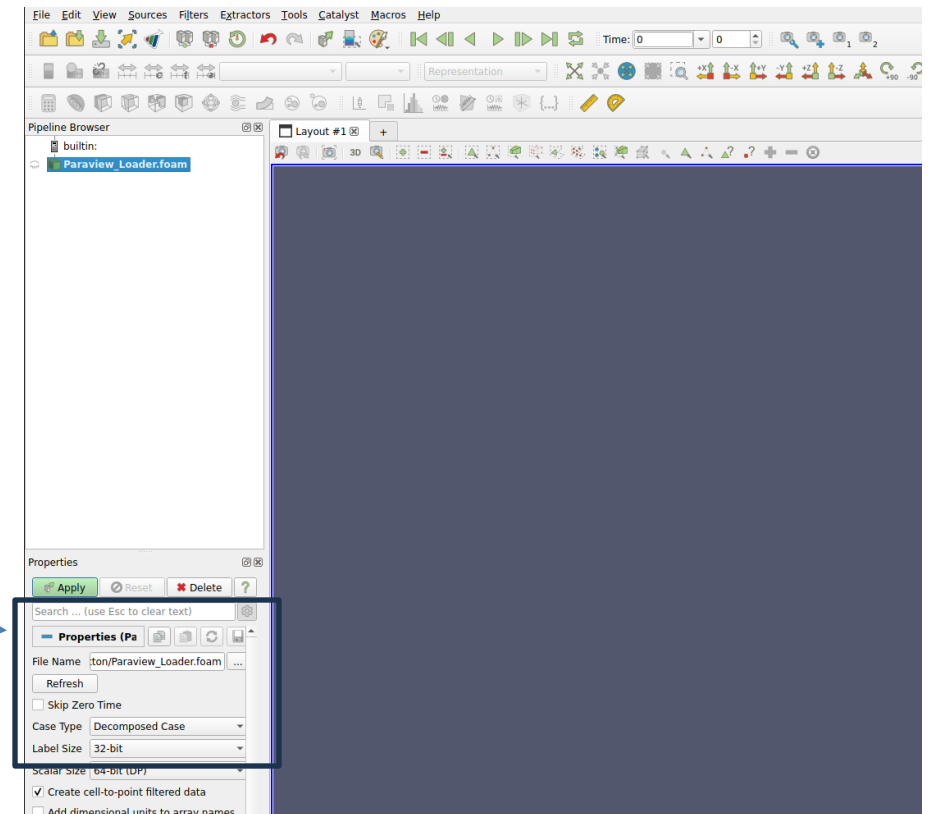
```
# Number of processors in each direction  
NPX=2  
NPY=2  
NPZ=2
```

Parallel decomposition – change according to how much core you can use!

Create the mesh and open paraview

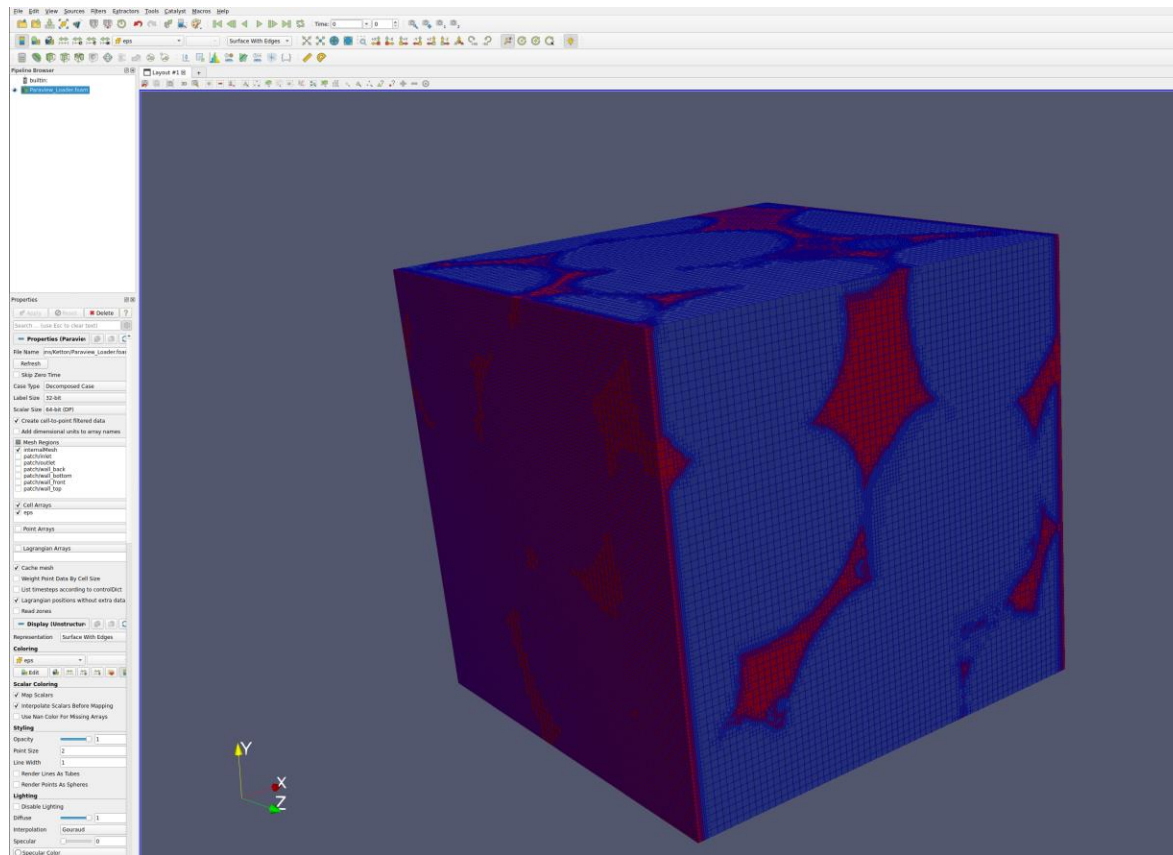
```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ ./createMesh.sh
create mesh
create eps
create topoSetDict
refine mesh
smooth solid surface
refine fluid/solid interface
processMeshCellCenters
create refined eps
smooth solid surface
refine fluid/solid interface
processMeshCellCenters
create refined eps
Mesh created. It is advised to check in paraview to confirm mesh and 0/eps are reasonable before running flow
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$
```

Unclick skip zero time and chose
"Decompose Case" (for a parallel case)

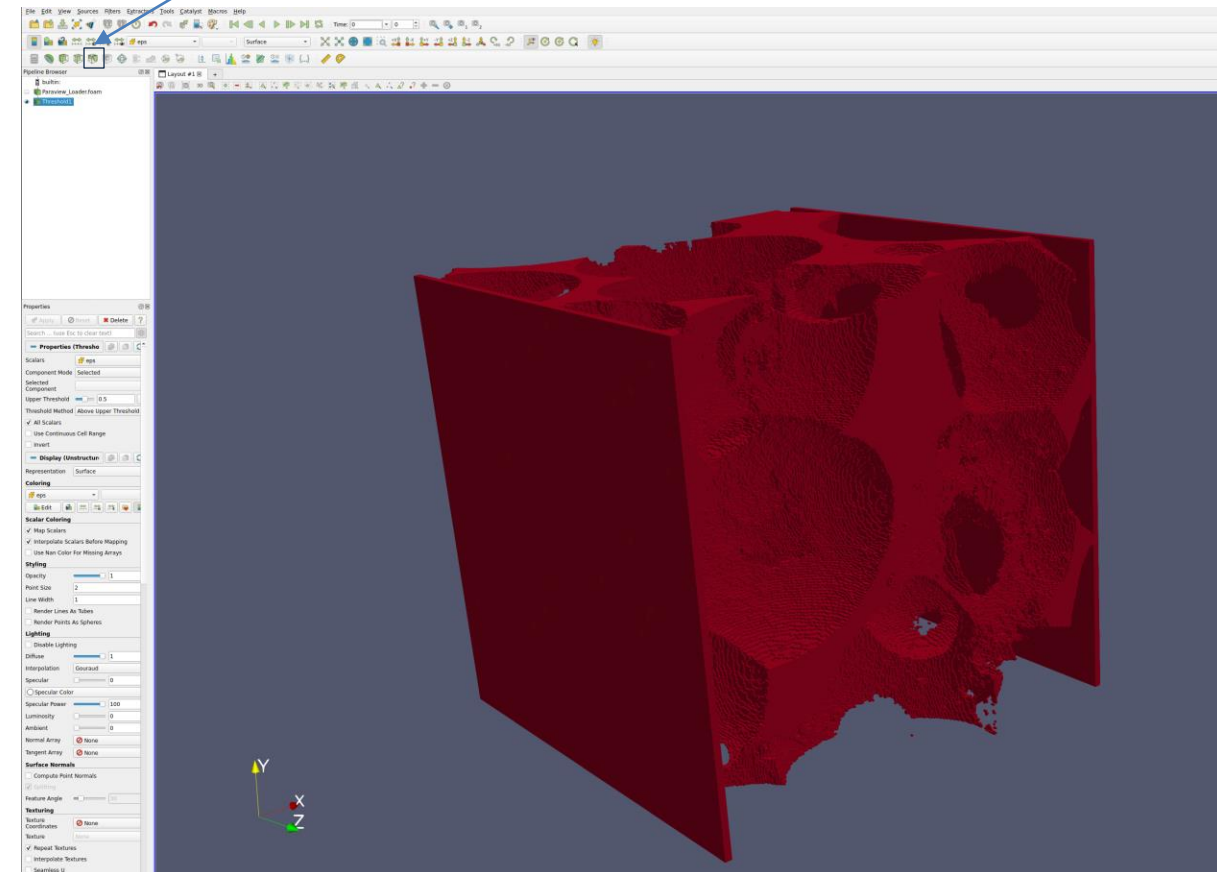


Use paraview to look at mesh

Select eps to see the solid (blue) and pores (red);
Choose "surface with edges" to see mesh



Chose threshold to keep cell eps>0.5



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InitCaseFlow.sh: flow and boundary properties

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ vi initCaseFlow.sh
```

```
#!/bin/bash

##### USERS INPUT #####

#Define boundary type
#boundary_type="flow_rate"
boundary_type="pressure_drop"
#Enter value (m^3/s) if boundary_type is flow_rate, otherwise put 0
flowRate=0 #4.2e-9
#Enter value (Pa) if boundary_type is pressure-drop, otherwise put 0
pressureDrop=0.1
#fluid kinematic viscosity (m2/s)
Visc=1e-6

model='grayscale'

#model='phases'

#define the labels of the phases
#phases=(1,2,3)

#define the porosity of each phase, note that the porosity of the solid phase CANNOT be 0, default to 0.0001
#micro_por=(1,0.35,0.0001)

#define the permeability (m2) of each label (note: solid phase should be < 1e-20, pore should be > 1e6)
#micro_k=(1e+12,1.82e-15,1e-26)

#Kozeny-Carman constant (m^(-2))
kf=1.8e12
```

Flow properties – keep for now

This means that Kozeny-Carman
will be used to penalise the solid
phase

$$K = 1 / (K_f * \epsilon_{\min}^3 / (1 - \epsilon_{\min}^2)) \sim e^{-25}$$

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ ./initCaseFlow.sh
```

Run the case and visualise flow field

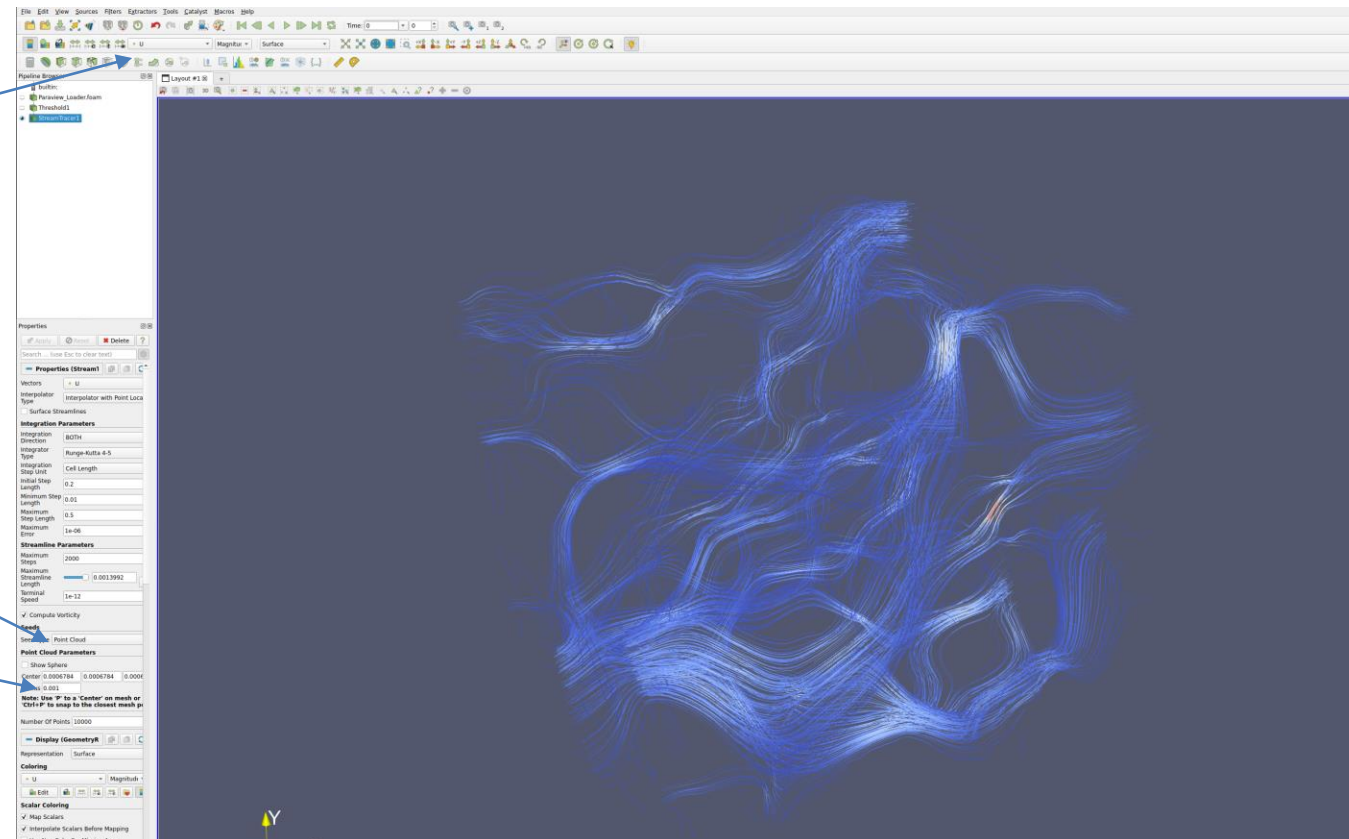
```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ time ./runCaseFlow.sh
Run simpleDBSFoam in parallel on 8 processors
Note: Please check the last line of simpleDBSFoamFlow.out to confirm the flow field has converged. If it has not, change TotalTime and/or the p tolerance in system/fvSolution and re-run the script
real    2m54.513s
user    22m27.859s
sys      0m43.134s
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ ./processFlow.sh
Run processPoroPerm in parallel on 8 processors
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$ cat poroPerm.csv
time poro perm Lpore Re UD tau
0 0.156361 8.53317e-12 9.03426e-05 0.357064 0.000617989 1.44124
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/Ketton$
```



Point cloud

Increase radius

10,000 seeds



Play time

- Change injection type to flow rate
- Increase mesh resolution
- Different image

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GeoChemFoam/Phreeqc

- Reactive transport modelling typically solve equations like:

$$\frac{\partial c_i}{\partial t} + \underbrace{\nabla \cdot (c_i u)}_{\text{Advection}} - \underbrace{\nabla \cdot (D \nabla c_i)}_{\text{Diffusion}} = \underbrace{f}_{\text{Chemical source}}$$

Equilibrium term, e.g. $a_{ij} \frac{\partial c_j}{\partial t}, a_{ik} \frac{\partial s_k}{\partial t}$

kinetic term, e.g. $k_{ij} c_j$

- GCFoam use Phreeqc for equilibrium term only
- Keyword supported are the ones related to SOLUTION and SURFACE

Operator Splitting

- The equations are solved using a sequential split-operator method
- First, GCFoam solve a non-reactive transport equation

$$\frac{c_i^{n+1/2} - c_i^n}{dt} + \nabla \cdot (c_i^{n+1/2} u) - \nabla \cdot (D \nabla c_i^{n+1/2}) = 0$$

- Then Phreeqc solve the chemical equilibrium

$$\frac{c_i^{n+1} - c_i^{n+1/2}}{dt} = f$$

- This splitting creates “operator splitting errors” since advection and diffusion are computed with intermediate $c_i^{n+1/2}$ state
- For error to be small, $CFL \leq 0.1 \rightarrow$ But slow!

Surface area

- For surface complexation, the equations are of the type:

$$\frac{\partial c}{\partial t} + a \frac{\partial s}{\partial t} = 0$$

- Surface concentration s :
 - Darcy-scale reservoir modelling: in kmol/kg (of rock): $a = \rho_R$
 - Pore-scale modelling: in kmol/m² (of surface): $a =$ specific surface area
- This is because at the Darcy-scale we can measure site density (kmol/kg) “easily”, but not surface area (unless micro-CT → pore-scale)
- At the pore-scale, surface area (of each voxel) is known

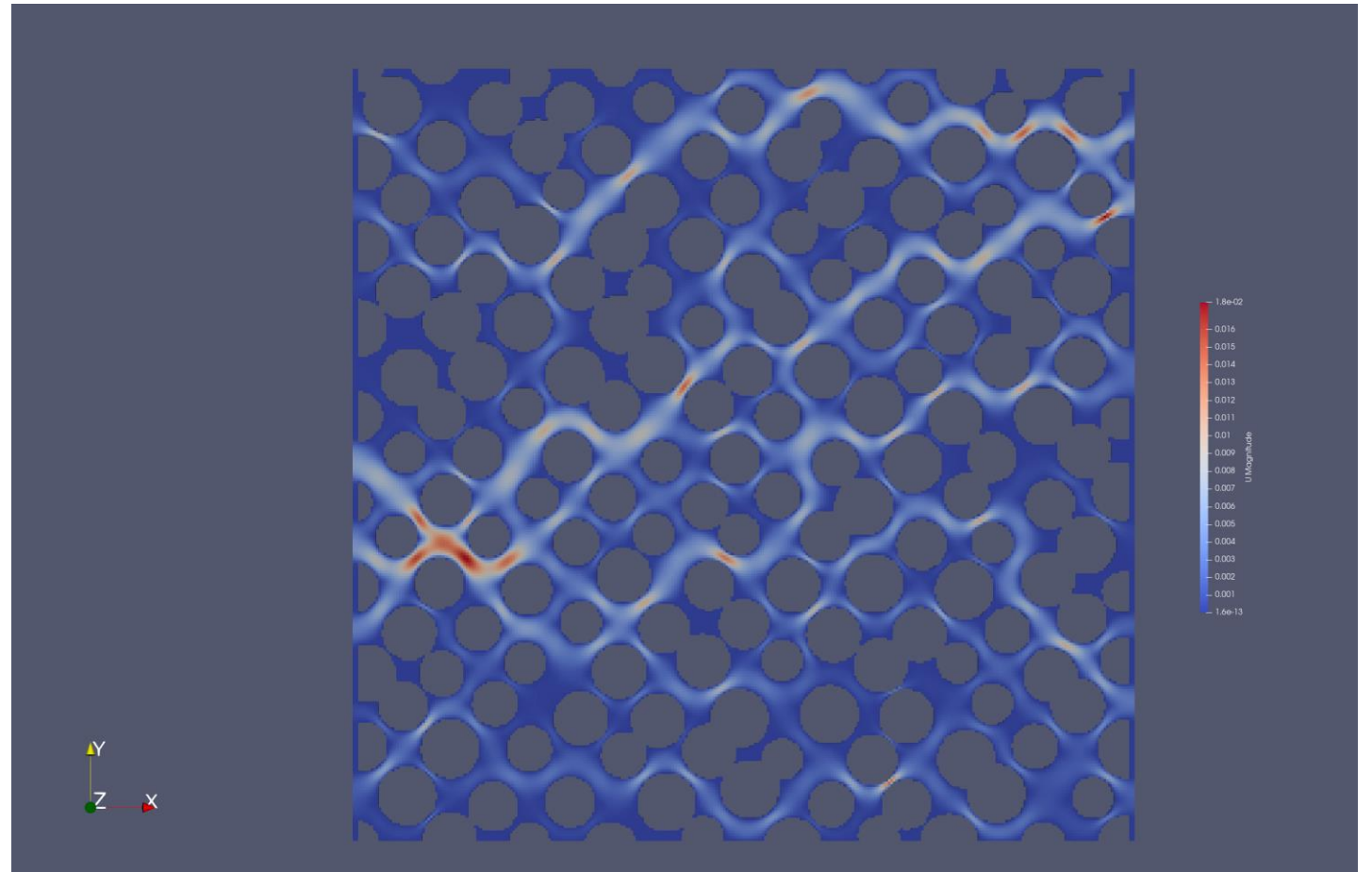
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Set-up the test case

```
(npython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs$ cp -r ../tutorials/reactiveTransport/reactiveTransportDBSFoam/HM12SolidGrainsSurfaceComplexation/ .  
(npython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs$ cd HM12SolidGrainsSurfaceComplexation/
```

Quickly check and run createMesh, initCaseFlow, runCaseFlow and processFlow and run them. These are similar to the case run this morning (except the image is 2D)



Check the phreeqcReactions file

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ vi constant/phreeqcReactions
```

```
TITLE Example 19.--Linear
```

```
SURFACE_MASTER_SPECIES  
  Surf_s Surf_sOH
```

```
SURFACE_SPECIES
```

```
  Surf_sOH = Surf_sOH
```

```
  Surf_sOH = Surf_sO- + H+
```

```
    -log_k -6.2
```

```
    -mole_balance Surf_sO-
```

```
  Surf_sOH + Ni+2 = Surf_sONi+ + H+
```

```
    -log_k 0.648
```

```
    -mole_balance Surf_sONi+
```

```
  Surf_sOH + Ni+2 + H2O = Surf_sONiOH + 2H+
```

```
    -log_k -7.91
```

```
    -mole_balance Surf_sONiOH
```

```
SOLUTION 0 formation water
```

```
  units  mol/L
```

```
  Ni      1e-13
```

```
  Cl      1e-13
```

```
  pH      7
```

```
SELECTED_OUTPUT
```

```
END
```

GCFoam currently only supports one surface "Surf" (may have more than one master species)

This solution does not have to be the initial one – it will be erased when GCFoam starts, but every required element needs to be present

Set-up the species in GCFoam with initCaseReactiveTransport.sh

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ vi initCaseReactiveTransport.sh
```

```
#!/bin/bash

##### USERS INPUT #####

#Species in the domain
species=("Ni+2","NiOH+","NiO2H2","Cl-","H+","OH-")

#Diffusion coefficients (m^2/s), one per species
Diff=(1e-9,1e-9,1e-9,1e-9,1e-9,1e-9)

#Initial concentration
C0=(9.838e-06,1.223e-07,9.722e-09,1.994e-05,1.003e-08,1.010e-06)
#inlet concentration
C_in=(0,0,0,0,9.974e-08,1.0095e-07)

#Surface master species and site density (kmol/m2)= 2e-3 mol/kg / 1 m^2/g
Surface_masters=("Surf_sO")
master_density=(2e-9)

#Surface species and initial concentrations (kmol/m2)
Surf_species=("Surf_sO-","Surf_sONi+","Surf_sOH","Surf_sONiOH")
Surf_C0=(4.631e-10,4.715e-10,8.997e-13,1.0644e-09)

# Minimum porosity included in the fixed PHREEQC chemistry mapping
eps_chemistry_min=1e-3

model='molDiff' ##Deff=Diff
## model='tortuosity'
## D=Diff/tau

#model='PecletDependent'
## Dipersivity constant
## De=1
```

Transported species

Initial concentration. This should be equilibrated – use Phreeqc!

Inlet concentration. Equilibrated?

Surface master and density

Surface species and initial concentration – Should be equilibrated with solution – use Phreeqc! Note unit: kmol/m²

Limit at which porosity is considered 0 for chemical reaction

Advanced tortuosity model – not used here

Run initCaseReactiveTransport.sh

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ ./initCaseReactiveTransport.sh
create concentration
Case initialised
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ vi runCaseReactiveTransport.sh
```

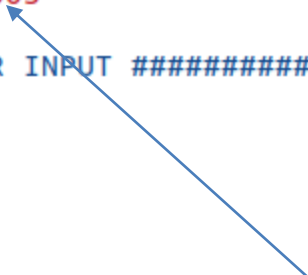
Open runCaseReactiveTransport.sh

```
#!/bin/bash

##### USERS INPUT #####

## Define the total time of the simulation and how often to output concentration fields
TotalTime=0.1
WriteTimestep=0.005
runTimestep=0.0005

#### END OF USER INPUT #####
```



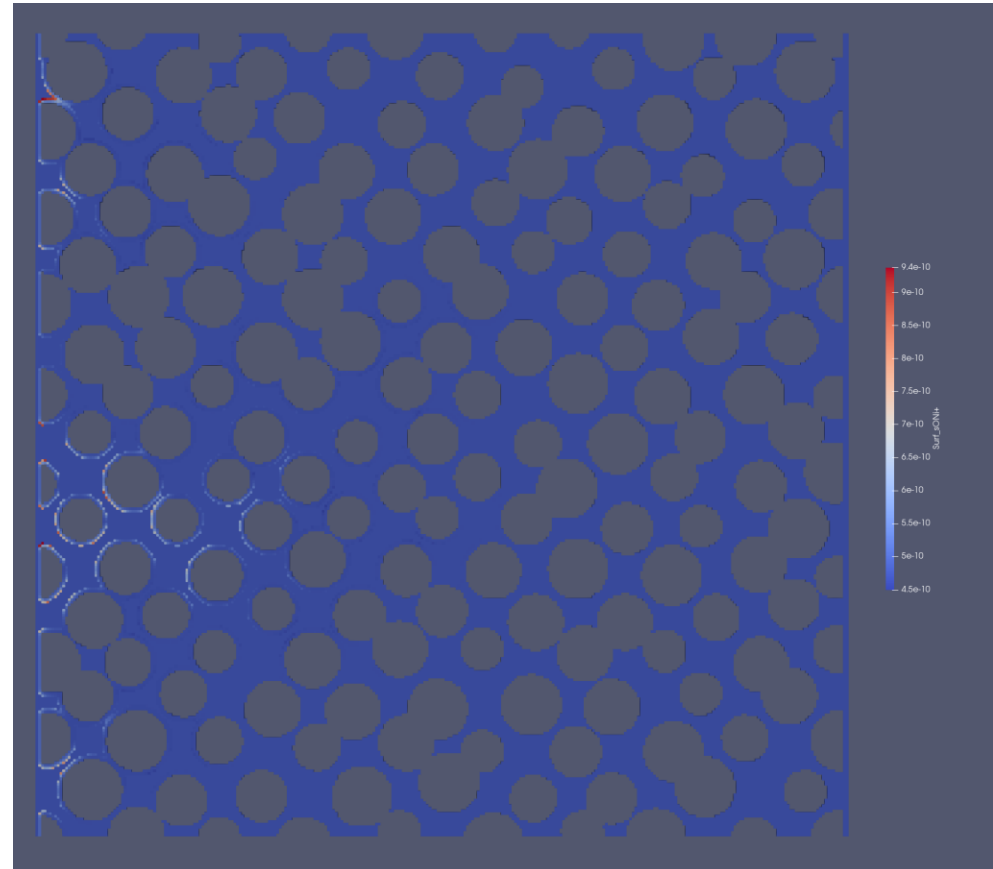
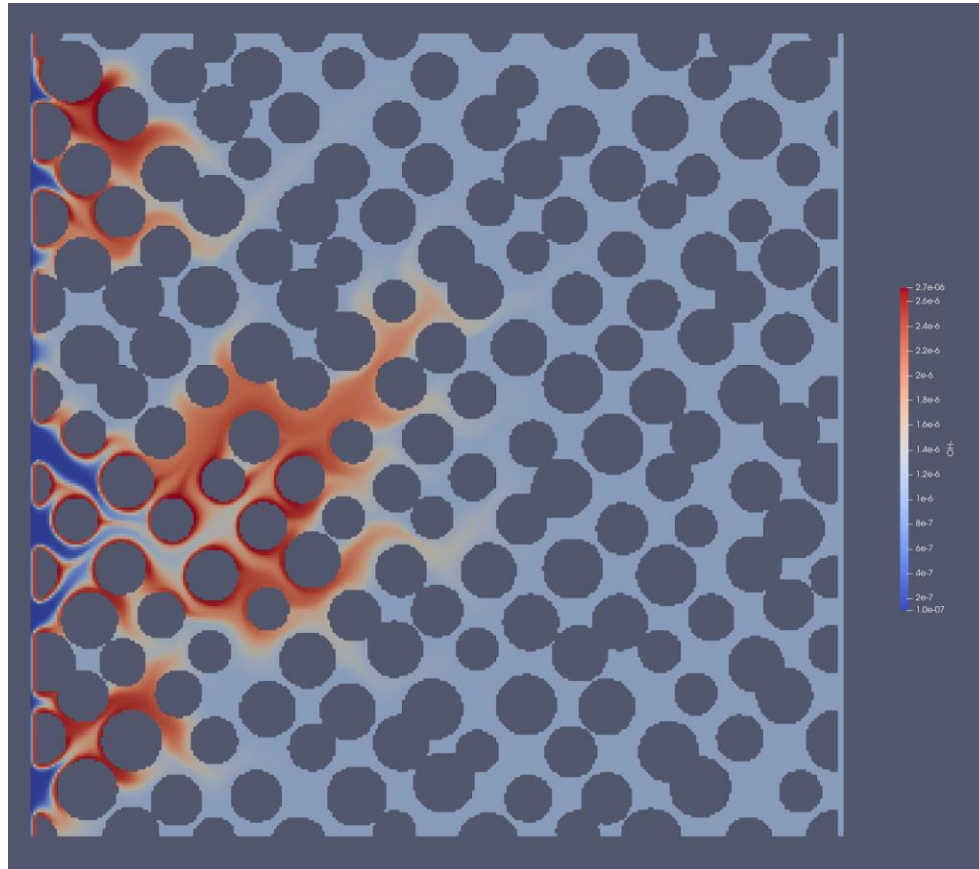
Time-step: reduce for accuracy, but longer compute time...

Run and look at the result

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ time ./runCaseReactiveTransport.sh
Run reactiveTransportFoam in parallel on 8 processors
```

```
real    2m17.220s
user    18m1.292s
sys      0m12.576s
```

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ █
```



Use processReactiveTransport.sh

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ ./processReactiveTransport.sh
Run processConcentration in parallel on 8 processors
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ cat Surf_s0
Surf_s0-_Conc.csv      Surf_s0H_Conc.csv      Surf_s0Ni+_Conc.csv      Surf_s0NiOH_Conc.csv
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ cat Surf_s0NiOH_Conc.csv
time S
0 1.0644e-09
0.005 1.06434e-09
0.01 1.06418e-09
0.015 1.06397e-09
0.02 1.06374e-09
0.025 1.06352e-09
0.03 1.06331e-09
0.035 1.0631e-09
0.04 1.06291e-09
0.045 1.06272e-09
0.05 1.06255e-09
0.055 1.06237e-09
0.06 1.06221e-09
0.065 1.06205e-09
0.07 1.06189e-09
0.075 1.06174e-09
0.08 1.06159e-09
0.085 1.06145e-09
0.09 1.06131e-09
0.095 1.06118e-09
0.1 1.06104e-09
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ cat H+_Conc.csv
time C
0 1.003e-08
0.005 1.03417e-08
0.01 1.06772e-08
0.015 1.07804e-08
0.02 1.07574e-08
0.025 1.06994e-08
0.03 1.06312e-08
0.035 1.05581e-08
0.04 1.04845e-08
0.045 1.04105e-08
0.05 1.03375e-08
0.055 1.02649e-08
0.06 1.01936e-08
0.065 1.01216e-08
0.07 1.00507e-08
0.075 9.97935e-09
0.08 9.90796e-09
0.085 9.83766e-09
0.09 9.76634e-09
0.095 9.69592e-09
0.1 9.62461e-09
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChenFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$
```

Use matlab to plot interesting curves

Agenda

- 9:30-10:00: Introduction to GeoChemFoam
- 10:00-11:00: From a micro-CT image to a GeoChemFoam case
- 11:00-11:15: Coffee break
- 11:15-12:15: Flow and permeability calculation
- 12:15-13:30: Lunch break
- 13:30-14:00: GeoChemFoam/Phreeqc coupling: how does it work
- 14:00-15:00: Reactive transport: Nickel (de)sorption on Na-Illite surfaces
- 15:00-15:15: Coffee break
- **15:15-16:15: Reactive transport: Nickel/Zinc competition on Na-Illite surfaces**
- 16:15-16:45: Conclusion and discussion

Clean the reactive part

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ ./deleteRT.sh  
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$
```

This will take the case back after the flow was computed – ready for
initCaseReactiveTransport.sh

Add zinc into constant/phreeqcReaction

```
TITLE Example 19.--Linear
SURFACE_MASTER_SPECIES
  Surf_s Surf_sOH
SURFACE_SPECIES
  Surf_sOH = Surf_sOH
  Surf_sOH = Surf_sO- + H+
    -log_k -6.2
    -mole_balance Surf_sO-
  Surf_sOH + Ni+2 = Surf_sONi+ + H+
    -log_k 0.648
    -mole_balance Surf_sONi+
  Surf_sOH + Ni+2 + H2O = Surf_sONiOH + 2H+
    -log_k -7.91
    -mole_balance Surf_sONiOH
  Surf_sOH + Zn+2 = Surf_sOZn+ + H+
    -log_k 2.068
    -mole_balance Surf_sOZn+
  Surf_sOH + Zn+2 + H2O = Surf_sOZnOH + 2H+
    -log_k -6.374
    -mole_balance Surf_sOZnOH

SOLUTION 0 formation water
  units  mol/L
  Ni      1e-13
  Cl      1e-13
  Zn      1e-13
  pH      7

SELECTED_OUTPUT
END
```

Don't forget to add zinc in the system

Modify initCaseReactiveTransport.sh

```
#!/bin/bash

##### USERS INPUT #####

#Species in the domain
species=("Ni+2","NiOH+","NiO2H2","Cl-","H+","OH-","Zn+2","ZnOH+","ZnO2H2")

#Diffusion coefficients (m^2/s), one per species
Diff=(1e-9,1e-9,1e-9,1e-9,1e-9,1e-9,1e-9,1e-9,1e-9)

#Initial concentration
C0=(9.838e-06,1.223e-07,9.722e-09,1.994e-05,1.003e-08,1.010e-06,0,0,0)
#inlet concentration
C_in=(4.919e-06,6.118e-08,4.862e-09,1.994e-05,1.003e-08,1.010e-06,4.472e-06,4.419e-07,7.007e-08)

#Surface master species and site density (kmol/m2)= 2e-3 mol/kg / 1 m^2/g
Surface_masters=("Surf_sO")
master_density=(2e-9)

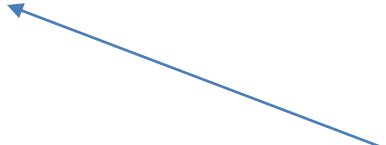
#Surface species and initial concentrations (kmol/m2)
Surf_species=("Surf_sO-","Surf_sONi+","Surf_sOH","Surf_sONiOH","Surf_sOZn+","Surf_sOZnOH")
Surf_C0=(4.631e-10,4.715e-10,8.997e-13,1.0644e-09,0,0)

# Minimum porosity included in the fixed PHREEQC chemistry mapping
eps_chemistry_min=1e-3

model='molDiff' ##Deff=Diff

## model='tortuosity'
## D=Diff/tau

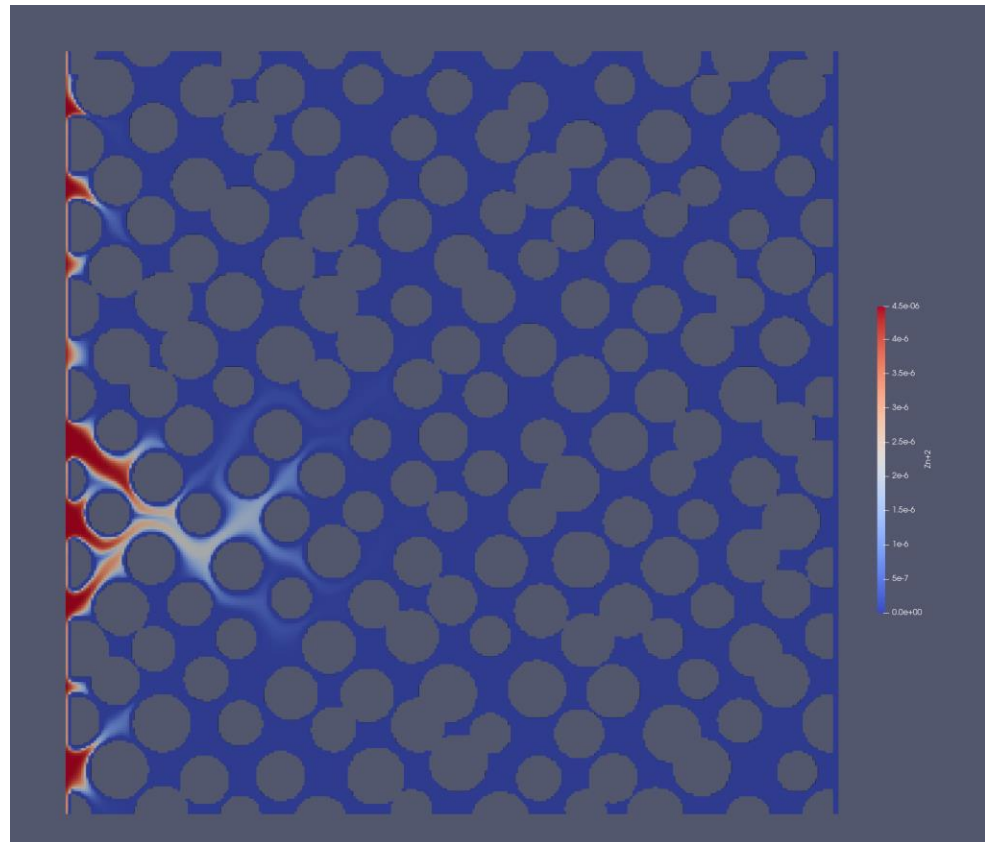
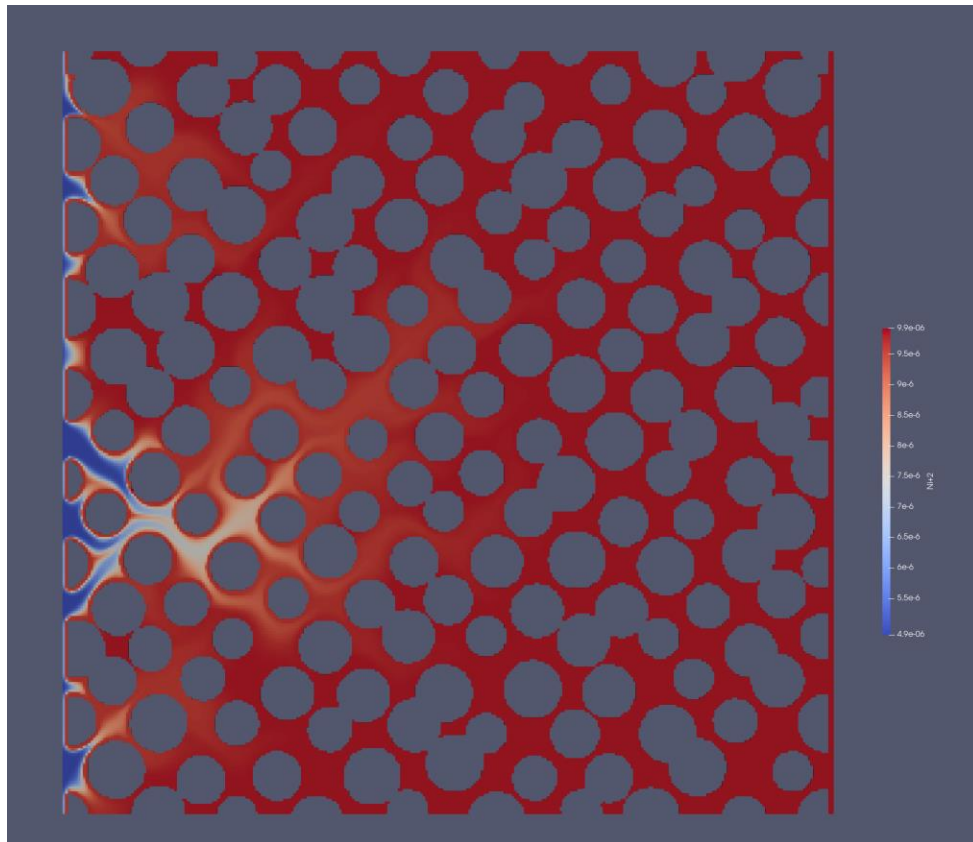
#model='PecletDependent'
## Dispersivity constant
## Pe<1
## Dx=Diff/tau*(1+betax*Pe^alpha1x)
## Dy=Diff/tau*(1+betay*Pe^alpha1y)
```



This solution was
equilibrated using
phreeqc

Run the case

```
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ ./initCaseReactiveTransport.sh
create concentration
Case initialised
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$ ./runCaseReactiveTransport.sh
Run reactiveTransportFoam in parallel on 8 processors
(mypython_environment) julien@julien-HP-EliteBook-855-G7-Notebook-PC:~/works/GeoChemFoam-5.2/runs/HM12SolidGrainsSurfaceComplexation$
```



Post-process the results

- What would be interesting to analyse?
- Nice animation?